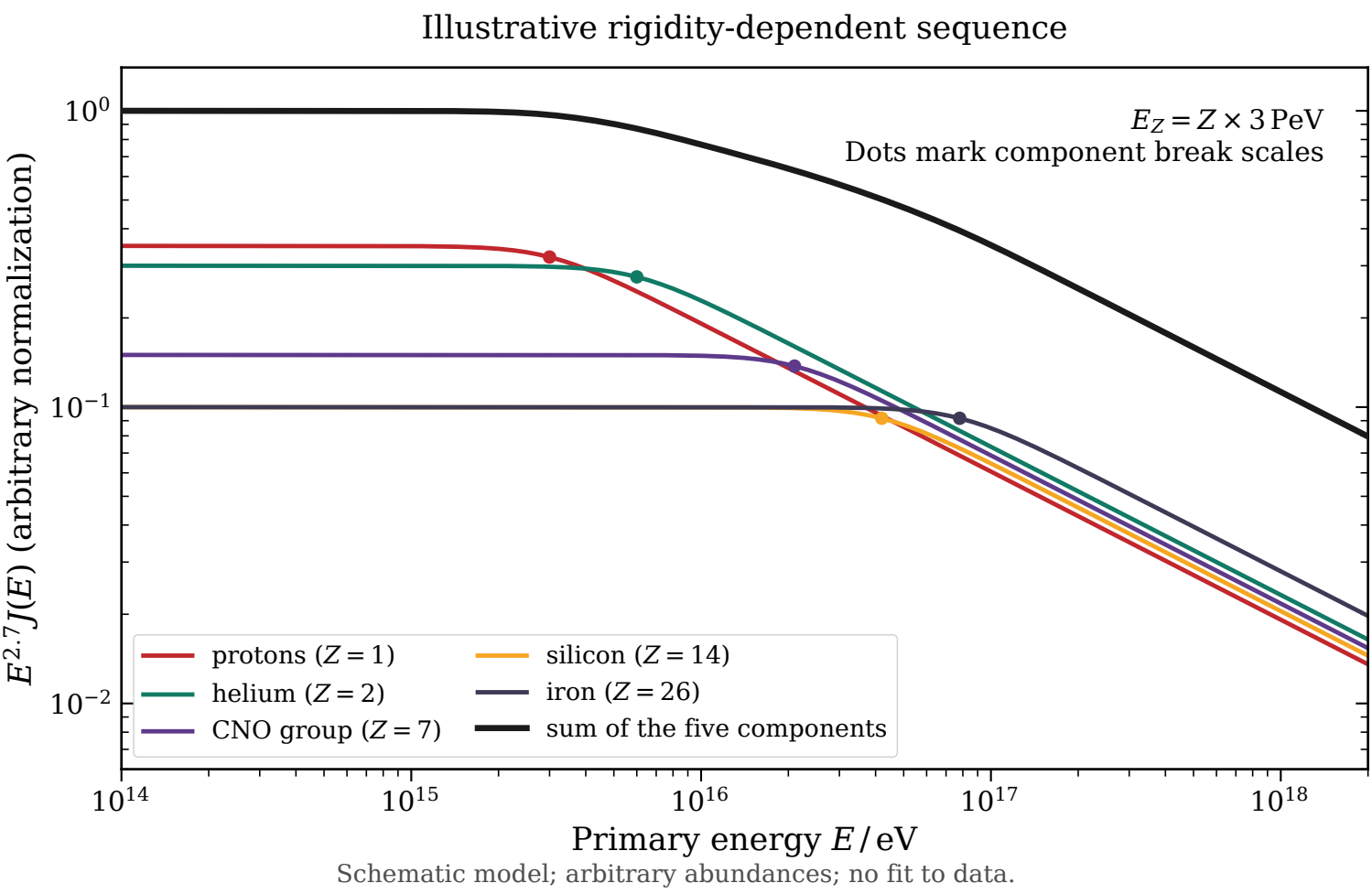
Illustrative rigidity-dependent sequence
$E_Z = Z \times 3\,\mathrm{PeV}$
Dots mark component break scales
$E^{2.7}J(E)$ (arbitrary normalization)
$10^{0}$
$10^{-1}$
$10^{-2}$
protons ($Z = 1$)
helium ($Z = 2$)
CNO group ($Z = 7$)
silicon ($Z = 14$)
iron ($Z = 26$)
sum of the five components
$10^{14}$
$10^{15}$
$10^{16}$
$10^{17}$
$10^{18}$
Primary energy $E$ / eV
Schematic model; arbitrary abundances; no fit to data.